Isaac Kanowski

isaackano1@gmail.com
https://orcid.org/0009-0000-2930-0566 github.com/isaackanowski

Education

PhD in Astronomy and Astrophysics Australian National University, Australia

Studying the impact of galaxy interactions at cosmic noon *2023 - Present*

- **Supervisors:**
  Assoc. Prof. Emily Wisnioski,
  Dr. Antoine Marchal, Dr. Takafumi Tsukui, Assoc. Prof. Luca Casagrande
- My PhD focuses on galaxy interactions at cosmic noon ($z \sim 1-3$) and their impact on galaxy star formation rates (SFRs) and turbulence.
- In my first project, I developed a spatially nonparametric kinematic deconvolution tool called `ROHSA-SNAPD` (Kanowski et al. 2025) by building upon the core features of an existing code `ROHSA` (Marchal et al. 2019). This python-based tool uses kinematic regularisation, which allows it to fit a kinematically diverse range of galaxies.
- I used `ROHSA-SNAPD` to model the kinematics of galaxies from the KMOS^{3D} survey (Wisnioski et al. 2015, 2019) in my second project. By crossmatching with the larger 3D-HST survey, I identified ~ 300 interacting and isolated galaxies in the KMOS^{3D} sample, which I used to calculate the interacting-driven changes in SFR and velocity dispersion. I found that interacting galaxies had higher SFRs, at a similar offset to interacting galaxies at $z \sim 0$. In contrast, I did not observe any increase in the velocity dispersion. This suggests that merger-driven velocity dispersion enhancement in the KMOS^{3D} sample is likely lower than in the local universe.
- I am currently conducting a follow-up study using photometric observations of edge-on discs to better understand the role of interactions on stellar disk thickness.

Bachelor of Advanced Science (Honours) Australian National University, Australia

Major: Astronomy and Astrophysics, minor: Mathematics *2019 - 2023*

- **Honours (first class):**
  Using strong gravitational lensing to constrain galaxy parameters,
  supervised by Dr. Trevor Mendel
- I measured baryonic + dark matter mass measurements of $z \sim 0.3$ early type galaxies which were gravitational lenses for distant ($z \sim 1-5$) Lyman-α emitters. I utilised `MUSE` IFU observations from the `MAGPI` survey (Foster et al. 2021) and the python pacakge `Lenstronomy` (Birrer and Amara, 2018; Birrer et al., 2021). I found generally good agreement with my lens masses and independent mass measurements from dynamical modelling by Derkenne et al. (2023) and stellar velocity dispersions within the same radii.

Publications

Spatially non-parametric recovery of intrinsic kinematic maps in pre- to post-merger galaxies

***Kanowski I.**, Wisnioski E., Mendel J. T., Marchal A., Tsukui T., 2025, RASTI, 4, rzaf040* *2025*

- In this work I introduced the spatially nonparametric deconvolution tool `ROHSA-SNAPD`. I demonstrated the tool's effectiveness by fitting to mock observations of both an idealised disc and an interacting system from a cosmological simulation.
- `ROHSA-SNAPD` is publicly released: github.com/isaackanowski/ROHSA_SNAPD

Do galaxy mergers increase star formation and turbulence at cosmic noon?

***Kanowski I.**, Mendel J. T., Wisnioski E., Förster Schreiber N. M., Marchal A., Tsukui T., (subm.)* *2026*

- In this publication I quantified the interaction-driven changes in star formation rates (SFRs) and velocity dispersions of a sample galaxies in the KMOS3D survey at cosmic noon. I found that the interacting sample had a similar increase in SFR to local galaxies, but had no observed increase in velocity dispersion. This suggests that interactions at cosmic noon do not substantially increase the level of turbulence in galaxies (less than a factor of 2).
- This paper is under peer review. A pre-print version is available on arXiv: https://arxiv.org/abs/2604.15002

CAIRO: Coherent Analysis of IFU data using Regularized Optimization

*Marchal A. et al. **(including Kanowski I.)**, (in prep)* *2026*

- This publication introduces a new kinematic fitting tool for IFU data.

Awards

Vice-Chancellor's HDR Travel Grant ($1500) *2026*

Joan Duffield Research Award ($2500) *2025*

Joan Duffield PhD Supplementary Scholarship ($5000) *2023*

Presentations

European Southern Observatory (ESO), Germany (presentation) *2026*

Max Planck Institute for Extraterrestrial Physics (MPE), Germany (presentation) *2026*

BeyondHubble conference, United Kingdom (poster) *2026*

University of Bath, United Kingdom (presentation) *2026*

Early Career Researcher Symposium - University of New South Wales, Australia (colloquium) *2026*

University of Sydney, Australia (presentation) *2026*

3D View of Galaxies online talk series (presentation) *2025*

ASTRO3D Science Meeting, Australia (presentation) *2024*

Teaching and mentoring

Co-Supervisor on undergraduate research project Australian National University, Australia

Co-Supervisor *2026*

- The project characterised the performance of the ROHSA-SNAPD tool on IFU data from the MUSE IFU instrument. I met weekly with the student, providing guidance on computing, data processing, and galaxy evolution background.

Honours and Masters mentoring program Australian National University, Australia

Mentor *2023 - 2024*

- I took part in a mentoring program to offer one-on-one support and advice to Honours and Masters students about their coursework and research.

First year introductory mathematics course Australian National University, Australia

Demonstrator *2022 - 2024*

- This introductory mathematics course covered a range of topics including trigonometry and differentiation.
- This role involved running weekly workshops where students worked in groups to solve course questions and complete quizzes. I helped students to work through problems and discussed possible solutions, aiming to help reinforce content taught in lectures.
- I also worked with other demonstrators to mark a number of course-wide assignments and exams.

First year mathematics course Australian National University, Australia

Demonstrator *2023*

- This mathematics course covered calculus and linear algebra.
- This role was similar to the mathematics demonstrating discussed above.

Joint International Science College Mathematics course Australian National University, Australia

Demonstrator *2023*

- This online short course was a collaboration between the Australian National University and Shandong University and focused on partial differential equations.
- This role involved running a series of weekly workshops over Zoom to help students with workshop questions. The nature of the Zoom workshops required encouraging students to communicate and work together effectively.

Youth Leading in STEM Mentor Australia

Tutor *2021*

- In this role, I was a mentor for high school students from across rural Australia. I drew from my own experience moving to university from a rural area.

Private Tutoring Australia

Tutor *2020 - 2022*

- In this role I privately tutored a high school student in mathematics, English, computing and conversation skills in order to prepare them for school and a future study.

Service

Weekly writing session Australian National University, Australia

Co-organiser *2025-Present*

- I help to organise weekly writing sessions for PhD students and academics. The aim is to create a focused environment conducive to writing.

Weekly journal club Australian National University, Australia

Co-organiser *2025*

- I organised a weekly journal club for PhD students and academics. This was a great opportunity to learn about a broad range of astronomy topics and practice presentation skills. Each week, someone would present and take questions about an interesting paper they had recently read.

Student welcoming committee Australian National University, Australia

Member *2025*

- I was part of the student welcoming committee in the astronomy department. This role involved contacting new PhD students before their arrival to provide information on transport and accommodation, and introducing them to the department and important contacts once they arrived.

National Masters/PhD student seminar Australian National University, Australia

Co-organiser *2023*

- I helped to organise a 3-day seminar for around 30 HDR students across Australia (https://nishamrutha.github.io/msss2023/).
- This role required working with a team of five other HDR students to advertise, organise catering and transport, schedule student presentations and invite academics to participate in advice sessions.

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